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The effect of integrated health management model on the health of older adults with diabetes in a randomized controlled trial

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ABSTRACT

The objective of this study was to evaluate the effect of integrated health management model on the health of older adults with diabetes. The 100 older adults with diabetes who gave informed consent were randomly allocated 1:1 into management and control groups. The integrated health management model was applied in the former while the latter was only given usual care. This model included the following components: health record establishment, health evaluation and health management (such as: diet advice, psychological aspects of health, education/skills training on health self-management, regular blood glucose monitoring, long-term diabetes drug monitoring, etc.). After 18 months, differences in three categories of variables (subjective grading items, objective measurement health indices and health service utilization) between the two groups before and after the intervention were assessed with *t*-test, χ^2-test and mixed model analysis. The management group demonstrated improvement on the following variables: health knowledge score, self-evaluated psychological conditions, overall self-evaluated health conditions, diet score, physical activity duration per week, regular blood sugar monitoring, waist-to-hip ratio, diastolic blood pressure and fasting blood sugar, the days of hospital admissions in the preceding 6 months. Mixed model analysis showed that gender, age, self-evaluated health status, self-evaluated psychological status, education level and resident status were important factors affecting health indices. This study demonstrated that integrated health management model was effectiveness in improving the health of older adults with diabetes.

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1. Introduction

With the development of society and economy around the world, the improvement of living standard, the changing of the dietary structure and the aging of population, the morbidity of diabetes is increasing year by year and has become a non-communicable chronic disease which is seriously harmful to the older adult health following after cardio-cerebrovascular diseases and malignancy. It brings heavy financial and social burden. New figures indicate that the number of people living with diabetes is expected to rise from 366 million in 2011 to 552 million by 2030 if no urgent action is taken (International Diabetes Federation, 2011). The prevalence of diabetes in China tends to increase in an incredible speed. It was estimated that the prevalence was 9.7% and the number was 92.4 million of adult diabetes aged 20 and over in China, the number of diabetics may rank the first in the word (Chinese Diabetes Association, 2011).

Diabetes itself is not fatal, but various complications caused by hyperglycaemia, such as ketoacidosis, cardio-cerebrovascular, diabetic nephropathy, will seriously influence the patients' survival and quality of life. Diabetes is closely related to life style. It cannot be controlled simply rely on traditional medical treatment, systemic management was required, such as maintaining good dietary habits, controlling weight, exercising regularly, obtaining regular foot and eye care, controlling blood glucose and complying with medications. American clinical guidelines also recommend that individuals with diabetes should tightly control glycemia, blood pressure and blood lipid levels in order to prevent acute complications and reduce the risk of long-term complications. Guidelines additionally recommend them to engage in regular physical activity, cease smoking and lose weight to those with obesity (American Diabetes Association, 2009). However, the study showed that recommended clinical preventive services were frequently below desirable levels (Gold, Devoe, Shah, & Chauvie, 2009). Even in developed country, about two-thirds diabetics could

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not manage effectively. It was serious in developing country, the result from the big or middle city of China in 2003, 2004, 2006 demonstrated that the HbA1c in only one quarter of diabetics reached the standard (<6.5%) (Chinese Diabetes Association, 2011). So it is significant to explore comprehensive interventions which can unite the force of society, hospital, family and individual.

Diabetic management could enhance the individual initiatives and realize this objective. There have been a number of studies of diabetic management (Drabik et al., 2012; Sacco, Bykowski, Mayhew, & White, 2012; Samoutis et al., 2010; Schlundt, Pichert, Gregory, & Davis, 2003). With a randomized controlled trail (RCT) study, Samoutis et al. (2010) showed that implementation of multifaceted quality improvement intervention program of diabetic patients in the primary health care settings could lead to the blood pressure, total cholesterol, low density lipoprotein-cholesterol, and three annual performance measures (urine protein testing, dilated eye and foot examination) improved at 18-month follow-up in the intervention as compared to the control group. Hirsch et al. (2002) suggested that in an academic family practice clinic, a multifaceted intervention in support of diabetes treatment guidelines modestly improved glycemic control without incurring additional costs. The improvement was mostly due to mitigation of the natural deterioration in control usually seen. Bruce, Davis, Cull, and Davis (2003) have demonstrated that the older patients confronted barriers to access or utilization of contemporary diabetes education, who are likely to benefit from specialized programs. It is important to study the education special for the older patients in order to improve the diabetic management. However there were few studies about these now.

In China, researches on diabetic's management have been carried out (Ruan et al., 2007; Sun, Liu, Wang, & Li, 2011; Tang, Yuan, & Tseng, 2005). Liu et al. concluded that the Chinese diabetes group visit model was a feasible, acceptable and effective alternative for supporting diabetes patient self-management in Chinese rural communities through a randomized controlled trial of 208 type 2 diabetes patients with aged 35–80 years. The model requires larger studies to determine its effect on blood glucose control and health care utilization (Liu et al., 2012). However most of the studies were done in the general people. The studies for the older adults with diabetes were few. Lu et al. implement the six months health management on 70 older adults with diabetes and showed that health management could improve the awareness rate of diabetes knowledge and proportion of regular monitoring, reduce the fasting and 2 h postprandial blood glucose, decline the glycosylated hemoglobin (HbA1c), blood pressure and BMI (Lu, 2008). But they only compared the results between the pre-intervention and post-intervention, not used the randomized controlled trials. Because of the few of randomized controlled trials for the management of the older adults with diabetes in China, the effectiveness of health management for older adults with diabetes has been inconclusive so far.

The purpose of this study was to evaluate the effect of the integrated health management model on the health of older adults with diabetes by using an RCT design.

2. Methods

2.1. Subjects

The outpatient of older adult with diabetes who signed informed consent forms and received clinic services between January and July in 2009 from a Nanjing district hospital endocrinology department were recruited after passing the inclusion and exclusion criteria stated below.

Criteria for the inclusion were: (1) local permanent resident; (2) aged 60 and over; (3) those diagnosed with type 2 diabetes according to the 1999 World Health Organization criteria. The exclusion criteria were: (1) cognitive defect, severe psychological disorder or psychological illness; (2) severe chronic diseases such as heart failure, respiratory failure, liver cirrhosis, renal failure or need for assistance in living; (3) limitations in physical activity; and (4) participating in or having participated in other trials within the last 30 days.

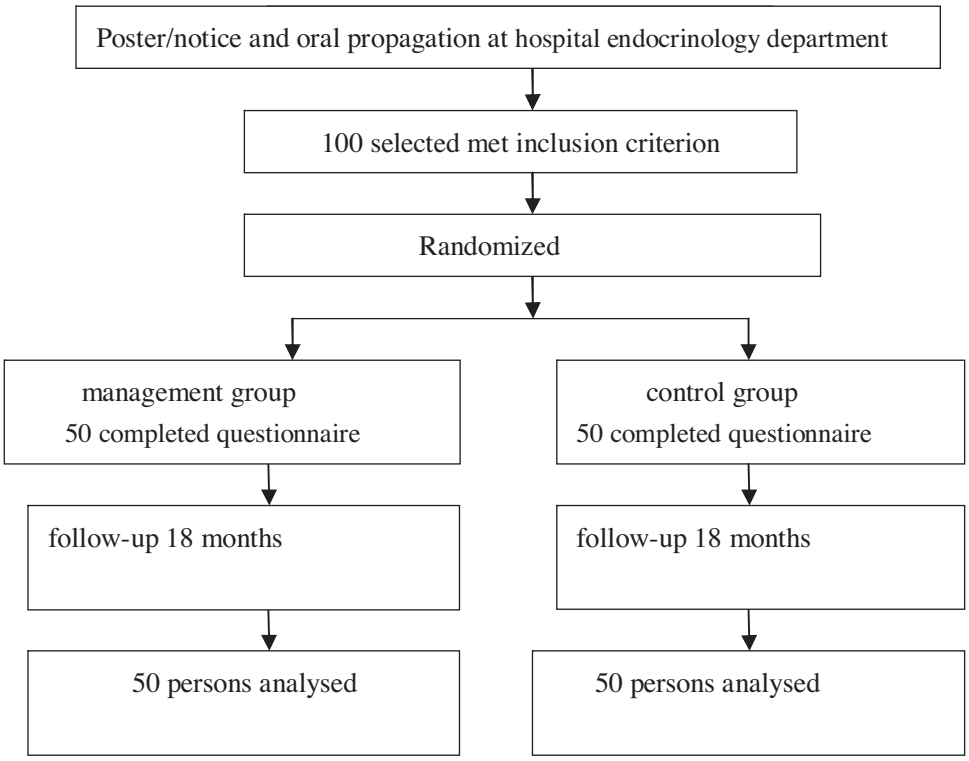

Fig. 1. Flow diagram of the progress in randomized controlled trial.

The 100 older adult with type 2 diabetes were randomly allocated to either the management or the control group at a 1:1 ratio using a random number table. This study had obtained the approval by the Medical Ethics Committee of Southeast University.

Flow diagram of the progress in randomized controlled trial is showed in Fig. 1.

2.2. Intervention

The management group received the integrated health management model including the following components: (1) Health record establishment, which was implemented by specifically-trained community health service center staff; (2) Health evaluation, which was done by related researchers through the health evaluation software; and (3) Health management, including diet advice, individual psychological aspects of health, a tailor-made exercise program based on an earlier evaluation, education/skills training on health self-management, individual telephone consultation, group lectures on health and diabetes (the prevention and treat for the diabetic foot disease, the cardio-cerebrovascular complications, the diabetic nephropathy, rational life style, self management, etc.), follow-up visits, regular blood glucose monitoring, long-term diabetes drug monitoring and distribution of health promoting materials, which were implemented by specifically-trained community health service center staff, managers and related researchers. The components of the intervention were 'administered' at least once per month by specifically trained community health service center staff, managers and related researchers (Fig. 2). The interventions and the compliance with the given advice were monitored. The management group intervention lasted for 18 months. The control group received usual care. After 18 months (December 2010), measurements were performed on both the management and the control groups.

2.3. Outcome measurements

At the end of the 18-month study, three different categories of health-related indices were assessed: subjective grading items, objective measurement health indices and health service utilization. Height and weight of the participants were measured (with a Model GRZ-120 human body weight gauge), and BMI was calculated according to the formula of weight/height^2. Waist and hip circumferences were measured according to the WHO MONICA guidelines. Blood pressure was measured using the brachial artery in the upper right arm with participants in the sitting position. A mercury sphygmomanometer was used. Blood pressure was measured three times and the average value was used as the outcome. The above indices were measured by a medically trained community physician. Fasting blood sugar and blood triglyceride were measured by the clinical test center of the hospital. Other indices such as diet, health knowledge score and physical activity duration were obtained by specially designed questionnaires. A questionnaire designed by the investigators was used to measure health-related indices. Before the full-scale formal investigation, a pilot investigation was carried out in some older adult diabetes and the questionnaire was modified according to the results of the pilot investigation. Any inconsistencies, such as missing or incorrectly recorded data, occurring in the investigation were checked and re-recorded again in time to ensure the quality of the investigational data.

2.4. Data analysis

The investigated data were input with Epidata 3.1 software (http://www.epidata.dk/) in a way of double-computer for double-person. The data quality was also controlled by setting up logic constraining conditions and valid value ranges of relevant variables. Before and after the intervention, related indices were compared between the management and the control groups with *t*-test (for continuous measurement data), χ^2-test (for categorical variables) performed in SPSS13.0 (SPSS Inc, Chicago, IL, USA), and mixed model (for factors affecting the health indices) performed by SAS version 9.1 (SAS Institute Inc., Cary, NC, USA.), with 0.05 set as the required level of significance.

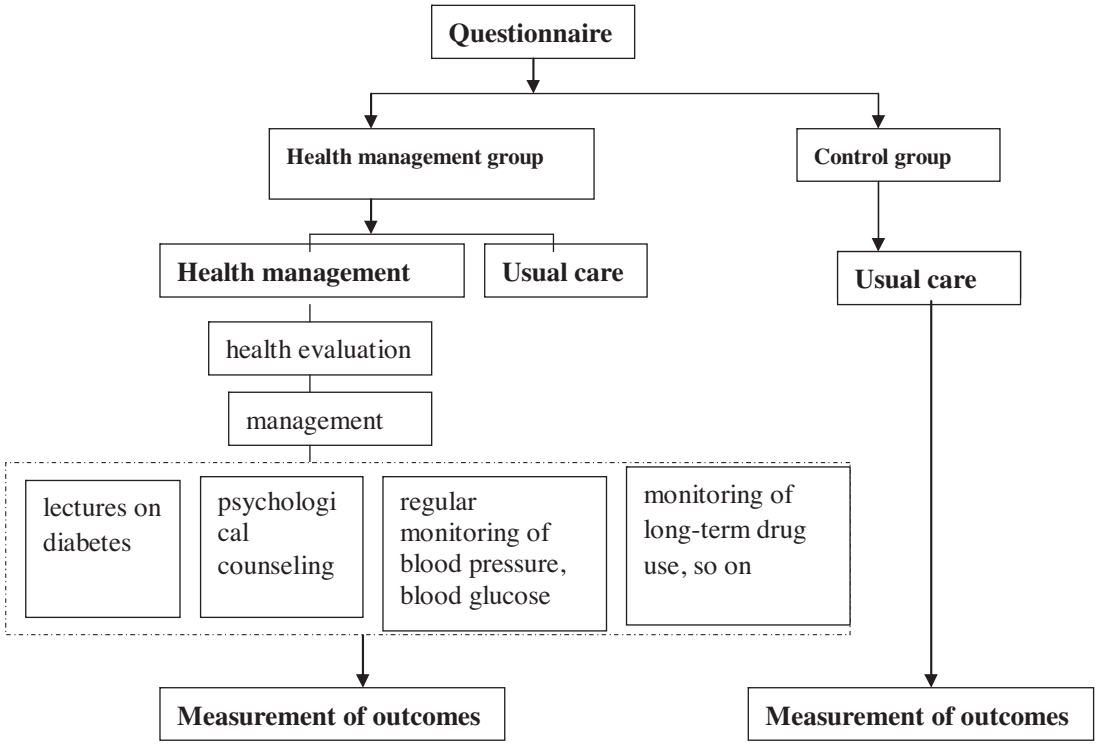

Fig. 2. Graphical depiction of the intervention.

3. Results

3.1. General characteristics of the subjects

Among the older adults with diabetes who received the baseline questionnaire, there were 50 persons in the management group and 50 persons in the control group. After 18 months, there was no one drop-out.

The general characteristics and health indices of the management and control group subjects at baseline were shown in Tables 1 and 2. The differences of general conditions and health indices between the management and the control groups were not statistically significant. This suggested that the two groups were balanced and comparable at the baseline level. In the management group, the mean age was 68.46 ± 5.99 years, 54% were males. 60% were husband and wife living together, 64% had a middle school education. Those 5 years and over of disease duration composed 44%, 64% had complications. In the control group, the mean age was 70.68 ± 6.83 years, 44% were males. 48% were living with their children, 42% had a middle school education. Those 5 years and over of disease duration composed 54%, 62% had complications.

3.2. Effect evaluation of the diabetic management

3.2.1. Comparison of health indices before and after intervention between the management and control groups

Changes in selected outcome measures following the 18-month intervention were presented in Table 2. Older adults with diabetes in the management group performed better than those in the control group on all subjective grading items (health knowledge score, self-evaluated psychological health status score, self-evaluated health status score, dietary score, physical activity duration per week, regular monitoring of blood sugar) ($P < 0.05$). Among the objective measured health indices, improvements in the management group were found in BMI, waist-to-hip ratio (WHR), diastolic pressure and fasting blood sugar compared with the control group ($P < 0.05$). For health service utilization, the days of hospital admissions in the preceding 6 months of management group reduced 1.74 days, while the control group increased 2.88 days, and there was statistically significant difference between the two groups ($P < 0.05$). However, the number of outpatient clinic visit increased 0.84 times in management group, the differences between the two groups was not statistically significant, the number of hospital admissions in the preceding 6 months was not different between the two groups after intervention.

Table 1
General characteristics between the management and control groups at baseline.

Variables	Management group (n = 50) n (%)	Control group (n = 50) n (%)	P
Age (years)			
Mean ± SD	68.5 ± 6.0	70.7 ± 6.8	0.189*
Range	60–81.0	60.0–81.0	
Gender			
Male	27 (54.0%)	22 (44%)	0.317
Female	23 (46%)	28 (56%)	
Resident status			
Married	36 (80.0%)	35 (77.8%)	0.799
Divorced or widowed	9 (20.0%)	10 (22.2%)	
Husband and wife living together	30 (60.0%)	20 (40%)	0.124
Living alone	5 (10%)	6 (12.0%)	
Living with children	15 (30%)	24 (48.0%)	
Education level			
Illiteracy	3 (6%)	11 (22.0%)	0.074
Primary school	11 (22%)	13 (26.0%)	
Junior high school	14 (28.0%)	12 (24.0%)	
Senior high school or Secondary school	18 (36%)	9 (18.0%)	
Junior college	4 (8.0%)	3 (6.0%)	
University and above	0 (0%)	2 (4.0%)	
Disease duration			
<1 year	7 (14%)	3 (6%)	0.542
~1 year	13 (26%)	12 (24%)	
~3 year	8 (16%)	8 (16%)	
≥5 year	22 (44%)	27 (54%)	
Complications			
Yes	32 (64%)	31 (62%)	0.836
No	18 (36%)	19 (38%)	

* t, the other χ^2.

3.2.2. Analysis of factors affecting on health-related indices

Mixed model was used to examine the effect of gender (X1), age (X2), education level (X3), marriage status (X4), resident status (X5), self-evaluated psychological status at baseline (X6) and self-evaluated health status at baseline (X7) on the before–after differences in the management and the control group. For the systolic blood pressure, it was significantly greater in association with the two factors: the age, those who are older among the respondents, and higher education level. For fasting blood glucose, the score was significantly greater for living with children. For physical activity, it was greatly influenced by self-evaluated health status at baseline, the difference was significantly greater for the worse at the baseline. For diet, it was significantly greater in association with the following factors: husband and wife living together, worse self-evaluated psychological status at baseline. In terms of WHR, the difference was significantly greater for males and the older (Table 3).

4. Discussion

This study evaluated the effects of the integrated health management model on the health of older adults with diabetes by using a randomized controlled trial. The results showed that the integrated health management improved several health indices of older adults with diabetes and decreased the days of hospital admissions during the 18-month study interval. Mu et al. (2012) also suggested that in order to curb the epidemic of type 2 diabetes in Asia, an integrated strategy employing a risk-based, holistic and, where necessary, multidisciplinary approach was needed.

Our subjective grading items included two parts: KAP and health practice. In KAP (knowledge-attitude-practice), the health knowledge score improved, the health attitude have improved in the self-evaluated psychological heath status and self-evaluated health status scores in management group. The older adults with diabetes have enhanced the consciousness of the responsible for their own health, and set up more positive and optimistic state of mind through health management. The health practice of management group has improved in following aspects: diet score, physical activity duration per week, the proportion of the monitoring blood sugar regularly. All of these suggested that health knowledge and attitude had produced promoting effect on the health behavior in certain, each part of KAP has mutual promotion and influence. These findings are consistent with the results reported by other studies (Davidson et al., 2009; Tessier et al., 2000). Davidson et al. (2009) showed in a randomized controlled trials that a combination of aerobic and resistance exercise improves physical fitness, glycemic control and insulin sensitivity in older adults and in people with diabetes. Al-Khawaldeha, Al-Hassanb, and Froelicher (2012) conducted a 223

Table 2
Comparison of health-related indices between management and control groups before and after intervention.

Variables	Baseline			After intervention		
	Management group Mean (SD) or *n* (%)	Control group Mean (SD) or *n* (%)	*P*	Management group the change Mean (SD) or *n* (%)	Control group the change Mean (SD) or *n* (%)	*P*
Subjective grade items						
Health knowledge scores (0–29)	8.74 ± 7.45	8.92 ± 6.67	0.899	13.72 ± 9.85	1.30 ± 8.27	0.000
Self-evaluated psychological health status score (1–5)	3.38 ± 0.75	3.54 ± 0.76	0.182	0.52 ± 0.99	0.18 ± 1.10	0.034
Self-evaluated health status score (1–5)	3.24 ± 0.74	3.28 ± 0.81	0.795	0.48 ± 0.91	0.00 ± 0.93	0.017
Diet (0–18)	10.52 ± 3.39	9.66 ± 3.44	0.179	5.40 ± 3.46	3.58 ± 3.62	0.012
Physical activity duration (minutes per week)	163.40 ± 350.09	157.50 ± 399.88	0.956	115.20 ± 402.18	18.10 ± 393.99	0.013
Regular monitoring blood pressure	29 (58.0%)	36 (72.0%)	0.142	37 (74.0%)	36 (72.0%)	0.822
Regular monitoring blood glucose	32 (64.0%)	40 (80.0%)	0.075	40 (80.0%)	38 (76.0%)	0.004
Objective measurement health indices						
Body mass index (BMI)	24.06 ± 3.00	24.32 ± 2.98	0.663	−1.58 ± 3.19	0.23 ± 2.91	0.004
Male BMI	24.11 ± 3.08	24.03 ± 2.38	0.924	−1.13 ± 3.77	0.78 ± 2.17	0.040
Female BMI	24.01 ± 2.97	24.55 ± 3.41	0.550	−2.11 ± 2.31	−0.20 ± 3.36	0.025
Waist-to-hip ratio (WHR)	0.89 ± 0.056	0.88 ± 0.055	0.294	−0.015 ± 0.64	0.021 ± 0.05	0.003
Male WHR	0.90 ± 0.065	0.88 ± 0.043	0.107	−0.01 ± 0.67	−0.01 ± 0.67	0.040
Female WHR	0.88 ± 0.037	0.88 ± 0.064	0.734	−0.03 ± 0.06	0.02 ± 0.06	0.020
Systolic blood pressure (mmHg)	133.30 ± 14.02	139.56 ± 18.72	0.098	−0.56 ± 12.51	0.18 ± 19.40	0.821
Diastolic blood pressure (mmHg)	82.28 ± 8.28	83.76 ± 8.81	0.647	−4.02 ± 11.45	0.60 ± 11.48	0.019
Fasting blood glucose (mmol/l)	7.89 ± 2.49	7.51 ± 2.40	0.435	−0.82 ± 1.90	0.06 ± 2.36	0.042
Health service utilization						
Number of outpatient clinic visits in the preceding 6 months	5.20 ± 6.25	5.64 ± 9.14	0.963	0.84 ± 6.13	0.20 ± 10.27	0.936
Number of admissions in the preceding 6 months	0.16 ± 0.47	0.10 ± 0.30	0.703	−0.080 ± 0.53	0.080 ± 0.53	0.133
The days of hospital admissions in the preceding 6 months	20.50 ± 7.32	10.00 ± 9.59	0.183	−1.74 ± 7.70	2.88 ± 9.93	0.049

diabetics aged 25 or older cross-sectional trial. The results demonstrated that the majority of subjects did not have their diabetes controlled, their self-efficacy was low and they had suboptimal self-management behaviors. Therefore, strategies to enhance and promote self-efficacy and self-management behaviors for patients are essential components of diabetes education programs. Furthermore, behavioral counseling and skill-building interventions are critical for the patients to become confident and be able to manage their diabetes. Bruce et al. (2003) have demonstrated significant associations between diabetes knowledge and diabetes education, dietetic advice and self-monitoring of blood glucose in subjects with type 2 diabetes. They suggested that

Table 3
Mixed model analysis of factors affecting health-related indices.

	Gender	Age	Education level	Resident status	Self-evaluated psychological status at baseline	Self-evaluated health status at baseline
	X1	X2	X3	X5	X6	X7
Systolic blood pressure						
Estimate		0.414	2.317			
SE		0.177	1.022			
P		0.021	0.026			
Fasting blood glucose						
Estimate				0.515		
SE				0.209		
P				0.016		
Physical activity						
Estimate						10.817
SE						3.687
P						0.001
Diet						
Estimate				−0.499	0.715	
SE				0.239	0.290	
P				0.039	0.016	
WHR						
Estimate	−0.024	0.002				
SE	0.010	0.001				
P	0.020	0.018				

Note: *SE* is standard error; *P* is probability.

each of these activities was important and a complementary source of information about diabetes and the elder patients confronted barriers to access or utilization of contemporary diabetes education, who are likely to benefit from specialized programs. Our diabetic management was specialized programs for the older adults with diabetes, so the effect may be related with these.

In the objective measurement health indices, improvements in the management group were found in BMI, waist-to-hip ratio (WHR), diastolic pressure and fasting blood sugar. Overweight and obesity are the common risk factors to cause chronic diseases which have been accepted by the most scholars through large number of studies. A comprehensive research on the main risk factors of the global burden of disease showed that chronic disease and injuries caused by the following seven risk factors had constituted the main burden of disease and health damage whether in developing or in developed countries: lacking of physical activities, unreasonable diet, dyslipidiemia, high blood pressure, overweight and obesity, smoking and drinking (Ezzati, Lopez, Rodgers, Vander, & Murray, 2002). Among these the weight controlling is the most important measure to reduce chronic disease. This study also showed that BMI, WHR in the management group were improved. Health management intervention can make patients master the relevant knowledge, get rid of bad life habits and improve the compliance of treatment; the encouragement and interaction between associates during the intervention, call visits played a supervision role, which could strengthen the patients' self-management consciousness; the practice of personalized health management intervention made the patients' condition controlled well. These findings were consistent with the other studies findings (Hirsch et al., 2002; Samoutis et al., 2010).

In terms of health service utilization, however, except the days of hospital admissions in the preceding 6 months, the number of outpatient clinic visit and the number of admissions did not change. It was most likely owing to the short length of the study. These findings are consistent with the results reported by Younmi, Kunsei, Eunyoung, Hyeongsu, and Kyung (2010). However, some studies showed that continuity of care, with a permanent physician–patient relationship, may be associated with decrease hospitalizations (Knight, Dowden, Worrall, Gadag, & Murphy, 2009) and emergency room visits (Gill, Mainous, & Nsereko, 2000; Ionescu-Ittu et al., 2007; McCusker et al., 2012).

In order to further analyze factors influencing the intervention effect, we also adopted mixed model to analyze the relationship between characteristics and the intervention effect. The mixed model improved the analysis of longitudinal data in which not only considering the relation for the same individual of different time, but also for the different individuals, with allocated groups as the factor between subjects and any variables which might influence outcome entered as covariates. Analyses showed that gender, age, self-evaluated health status, self-evaluated psychological status, education level and resident status were important factors affecting health indices (subjective and objective). The study of Huang et al. on the older adult showed the similar result (Huang, Chen, Yu, Chen, & Lin, 2002). More specifically, self-evaluated health conditions and self evaluated psychological status were positively affected by health management, the worse at the baseline the greater possibility we could improve them through health management, which improved the cognition, slow down in progression of the underlying disease, raised the confidence and positive attitude towards a healthy life style. In summary, the measures for specific factors should be taken to increase the effect of diabetic management during the implementation of diabetic management.

There are a number of limitations in our trial. Although our research has demonstrated that health management can improve the health of the older adult in several categories (subjective grading items, objective measurement health indices, health service utilization and multiple regression analysis on factors affecting the health index differences), some long-term indices such as mortality have not yet been obtained and the number of outpatient clinic visit, the number of admissions between the two groups was not significantly different. This most likely was due to the short duration of the study (only 18 months of health management intervention). In addition, the limitations of our study included the small sample size. In the future, extending the duration of the study could be helpful for further investigation of the long-term effects of health management on the health of the older adults with diabetes, extending the sample is also needed.

5. Conclusions

With a randomized controlled trial set in Nanjing, China, we have demonstrated that the integrated health management could improve the health of the older adults with diabetes. This is true both for scores of subjective grading (health knowledge score, self-evaluated psychological status, self-evaluated health status, dietary score, physical activity duration per week, regular monitoring of blood sugar) and for objective measurement health indices (BMI, WHR, diastolic pressure, and fasting blood sugar), as well as for health service utilization (the days of hospital admissions in the preceding 6 months).

Conflicts of interest

None.

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